

DIVISION S-5—SOIL GENESIS, MORPHOLOGY & CLASSIFICATION

Micromorphological Evidence of Polygenesis of Three Aridisols, Western Mojave Desert, California

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ABSTRACT

Soils in arid regions are often polygenetically related to climatic variation. On a dissected alluvial fan in the western Mojave Desert, a Typic Durorthid (loamy, mixed, thermic, shallow; Alko series) and two Typic Haplargids (fine-loamy, mixed, thermic; Neuralia and Garlock series) were selected for micromorphological investigation to determine the sequence of pedogenic events. The micromorphology of pedogenic silica, carbonate, and clay is described for duripans, duric horizons (Btqk), and underlying buried Bt horizons in these soils. In the Alko and Neuralia soils, rodent bioturbation has transported sub-surface calcareous fragments, thereby enriching the soil surface with silica and CaCO_3 . These fragments are not completely removed from the surface by leaching under the present-day climate. Clay illuviation and silicification occurred simultaneously in the lower duripan (40–150 cm) of the Alko, with subsequent accumulation of CaCO_3 in seams and root channels. Silicification of the Btqk and buried Bt horizons in the Neuralia and Garlock soils occurred after clay illuviation. The complete cementation of the lower duripan in the Alko soil has apparently isolated the buried Bt, but the buried Bt horizons in the Neuralia and Garlock soils have undergone further calcification and silicification. We speculate that the polygenesis of soils in the study area is related to climatic fluctuations since the mid-Pleistocene and to bioturbation of surface horizons. Development of carbonate-free argillic horizons probably occurred during pluvial periods, whereas calcification occurred during drier periods. Silicification appears to have been contemporaneous with both clay illuviation and calcification and, thus, may be related to pedochemical conditions rather than to climate.

THE ROLE OF MICROMORPHOLOGY in geomorphology has been reviewed by Mücher and Morozova (1983), who described how micromorphological observations may be used to characterize and interpret Pleistocene landscape history. The effectiveness and limitations of soil micromorphology for obtaining paleoenvironmental information from paleosols are described by Fedoroff et al. (1990). Nettleton et al. (1990) studied the micromorphology of the argillic horizons in Haplargids and Argiborolls in southwestern Colorado and concluded that they have formed in a climate similar to that of the present. Gile et al. (1966) examined the carbonate morphology of Aridisols in New Mexico and concluded that polygenesis of some Aridisols is related to climate.

Calcareous soils with duripans and duric horizons are widespread in the arid southwest USA (Flach et

al., 1974; Chadwick et al., 1987; Boettinger and Southard, 1990). Soils with a duripan have often developed in volcanic materials or have been covered by volcanic tephra during their pedogenesis (Soil Survey Staff, 1975; Chadwick et al., 1989). Duripans have also formed in soils that have no trace of volcanic glass; their silica cement is apparently derived from weathering of silicate minerals, such as feldspars (Torrent et al., 1980; Boettinger and Southard, 1991).

Aridisols with duripans and duric horizons occur in deposits estimated to range in age from 100 to 500 ka on dissected alluvial fans in the western Mojave Desert (Eghbal and Southard, 1993a). Variation in the distribution of CaCO_3 , clay, and silica in these soils suggests that they have been influenced by different sets of soil-forming processes during their pedogenic history. Micromorphological investigation of polygenetic soils on deposits of different ages may allow reconstruction of the sequence of past events. Therefore, the objectives of this study were to: (i) describe the micromorphological relationships of pedogenic silica, carbonate, and clay in duripans and duric horizons of Durorthids and Haplargids in the study area, and (ii) examine the implications of silica, carbonate, and clay distributions for the sequence of pedogenic events.

MATERIALS AND METHODS

Study Site

The study area is in the western Mojave Desert (35°10'N, 117°55'W) on a west-facing dissected alluvial fan of the Rand Mountains (Fig. 1), composed primarily of Mesozoic granitic rocks. The alluvial fans in the study area are derived from materials originating on deeply weathered granitic pediments. These fans have 1 to 2% regional slope. The dissected parts of the fans show a bar and swale topography in the aerial photographs. The climate is warm and arid with a mean annual air temperature of 18 °C and mean annual precipitation of ≈110 mm, falling between November and March (Valverde and Hill, 1981). The vegetation is dominated by creosote bush [*Larrea tridentata* (Sessé & Moçino ex DC.) Cov] and fiddleneck [*Amsinckia tessellata* Gray]. Four soils were mapped on the dissected alluvial fans; the Neuralia and the Alko series are dominant, with inclusions of the Cajon (mixed, thermic Typic Torripsamment) and the Garlock soils (Valverde and Hill, 1981).

Soils

The Alko, Neuralia, and Garlock soils were exposed in a trench excavated from an interfluvium toward an ephemeral stream channel on a dissected alluvial fan (Fig. 2). Detailed study of

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Abbreviations: BSE, back-scattered electron; EDXRA, energy dispersive x-ray analysis; ka, thousand years; SEM, scanning electron microscope; XRD, x-ray diffraction; ICP, inductively coupled plasma.

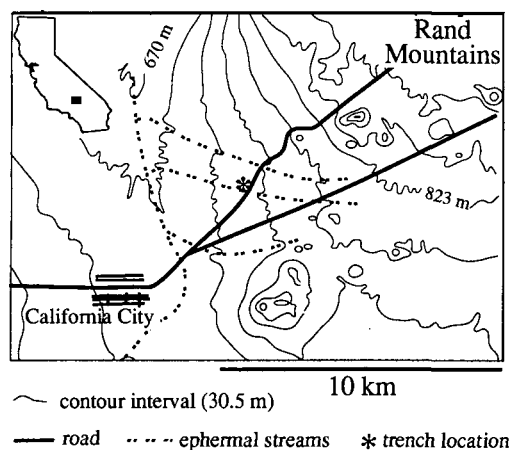


Fig. 1. Location of the study site in the western Mojave Desert.

soils in the area showed that Alko occurs on the edges of broad summits and on the crests of narrow summits, Neuralia (calcareous throughout) on the top of broad summits and the sideslopes, and Garlock (noncalcareous to 65 cm) on the sideslopes. Details of field and stratigraphic relations are presented by Eghbal and Southard (1993a).

The Alko soil, in the first 10 m of the trench from the summit, has a laminar-capped, calcareous duripan at a depth of 10 to 25 cm and another less calcareous and morphologically different duripan from 40 to 150 cm (Table 1). The upper duripan of the Alko soil is not continuous, but occurs as platy fragments in the profile. The middle of the trench, 10 to 26 m from the summit, exposes the Neuralia soil, which is calcareous throughout and has no duripan (Fig. 2). A duric horizon (Btqk) is found between the depths of 65 and 145 cm. Krotovinas were observed in the profiles of the Alko and Neuralia soils. Rodents apparently bring pieces of calcareous soils

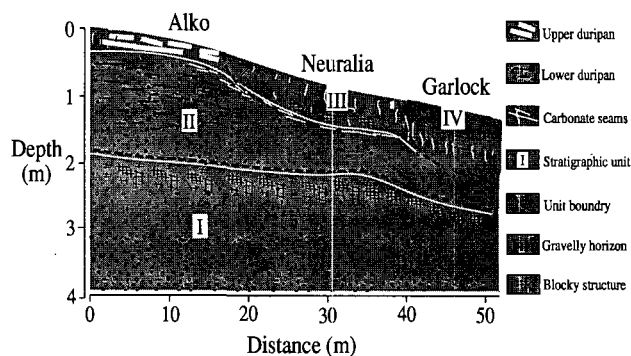


Fig. 2. Stratigraphy of soils in the excavated trench from an interfluvial toward an ephemeral stream channel (vertical exaggeration).

and duripan fragments to the surface as they burrow through these soils. Both Alko and Neuralia soils are calcareous to the surface and the distribution of CaCO_3 has been affected by bioturbation. The Garlock soil, exposed in the trench on the toe slope, is noncalcareous to a depth of 65 cm and has an argillic horizon, with a 5YR hue and coarse prismatic structure (Fig. 2). No evidence of significant bioturbation was observed in the Garlock soil. A Btqk horizon, similar to the Btqk of the Neuralia soil, occurs at a depth of 80 to 150 cm (Table 1). A buried soil underlies the three soils and was exposed in the trench at a depth of 1.5 to 2 m.

Four alluvial units were observed in the trench (Fig. 2): (I) the deposits in which the buried soil formed, (II) the sediments overlying the buried Bt horizon in which the Alko soil formed, (III) sediments in which the Neuralia soil on the sideslope occurs, and (IV) sediments in which the Garlock soil formed. Paleomagnetic analysis of Unit I showed that it has reversed polarity and is probably older than 783 ka (Bakshi et al., 1992). Although no radiometric data are available, based on the soil

Table 1. Selected morphological, physical, and chemical properties for representative horizons of the three soils in the excavated trench.

Depth cm	Horizon	Moist color	Texture†	pH	Gravel‡	Clay	CaCO_3	SiO_2	Al_2O_3	$\text{SiO}_2/\text{Al}_2\text{O}_3$
% g kg ⁻¹ <2 mm										
Alko, Typic Durorthid										
0–1	A1	10YR 5/6	grcosl	8.2	15	113	41	14	6	2.3
1–10	A2	10YR 5/4	sl	8.7	5	225	78	22	11	2.0
10–12§	Bqkm	10YR 8/2	—	—	—	—	742	38	4	9.5
12–25	Bqk	10YR 5/6	cosl	8.6	—	133	114	36	10	3.6
40–100	Btqkm	7.5YR 4/6	grcosl	8.8	20	38	6	28	9	3.1
190–215	Btqkb	7.5YR 4/4	scl	8.6	—	294	2	35	16	2.2
Neuralia, Typic Haplargid										
0–1	A1	10YR 5/4	grcosl	8.1	30	107	13	13	6	2.2
1–10	A2	10YR 5/4	sl	8.5	5	105	11	15	8	1.9
30–65¶	Bk	10YR 5/6	sl	8.7	13	166	24	13	7	1.9
30–65¶	Btk	7.5YR 5/6	scl	8.7	10	208	31	15	9	1.7
80–115	Btqk	7.5YR 4/4	grlcos	8.7	33	109	12	17	8	2.1
145–160	Btqkb	7.5YR 5/4	scl	8.7	8	202	11	31	14	2.2
Garlock, Typic Haplargid										
0–1	A1	10YR 4/4	grlcos	8.2	15	100	2	10	6	1.7
1–6	A2	10YR 4/4	sl	8.4	3	133	1	11	8	1.4
50–65	Bt	5YR 4/4	scl	8.6	—	237	0	19	15	1.3
135–150	Btqk	7.5YR 5/6	vgrsl	8.3	50	161	4	41	14	2.9
150–180	Btqkb	7.5YR 5/4	cosl	7.9	3	116	66	46	12	3.8

† Texture: cosl = coarse sandy loam, grcosl = gravelly coarse sandy loam, grlcos = gravelly loamy coarse sand, scl = sandy clay loam, sl = sandy loam, vgrsl = very gravelly sandy loam.

‡ Field-estimated volume.

§ The Bqkm horizon is the laminar cap of the upper duripan.

¶ The Bk and Btk portions of the Bk/Btk horizon (30–65 cm) were sampled and analyzed separately.

development, stratigraphic limitations, and correlation with other apparently climatically controlled units in the Mojave Desert, Units II, III, and IV are estimated to be >300, 200, and 100 ka, respectively (Eghbal and Southard, 1993a).

Laboratory Analyses

Undisturbed ped and clod samples were impregnated with Petropoxy-154 resin (Palouse Petro Products, Palouse, WA) for thin-section preparation. Thin sections were also prepared from samples of the calcareous duripan of the Alko soil after treatment with 2 M HCl to remove CaCO_3 . From one such sample, as many of the mineral grains of quartz and feldspars as possible were separated under a dissecting microscope and the mineralogy of the remaining silica cement was analyzed by a Diano XRD 8000 x-ray diffractometer (Diano Corp., Woburn, MA).

The surfaces of a few thin sections were polished and coated with C for BSE imaging using a Cameca Sx-50 microprobe (Cameca, Paris, France). Cemented grains were coated with Au-Pd alloy and observed with a Hitachi SEM (Hitachi Scientific Instruments, Mountain View, CA). Qualitative analysis of selected morphological features was performed by EDXRA, collecting x-ray spectra for 20 s using a Tracor Northern (EDXRA detector (Tracor Northern, Middleton, WI).

Particle-size distribution was determined using the pipette method (Gee and Bauder, 1986), following removal of organic matter with NaOCl (commercial chlorine bleach) and of soluble salts and CaCO_3 with pH 5.0 NaOAc (Jackson, 1979). Total CaCO_3 equivalent was determined on <180- μm ground samples by a manometric measurement of evolved CO_2 (Soil Conservation Service, 1984). Bulk density of laminar duripans was measured using the paraffin clod method as described by Singer (1986). Noncrystalline and short-range-ordered aluminosilicates were extracted, after citrate-bicarbonate-dithionite treatment, by boiling 0.5 to 1 g of <180- μm oven-dried, ground samples in 0.5 M NaOH (Jackson et al., 1986). Silicon and Al were quantified on an ARL-3510 ICP spectrometer (Applied Research Laboratories, Sunland, CA).

RESULTS AND DISCUSSION

Calcareous Laminar Duripan in the Alko Soil

The upper duripan (10–25 cm) of the Alko soil has a laminar cap that consists of two parts. A calcareous layer 1 to 2 cm thick, containing spherical nodules 0.2 to 1 mm in diameter, normally lies on top of a much harder silica-rich layer 1 to 3 mm thick (Fig. 3A and 3B). The laminar cap meets the requirements for a duripan and remains intact after treatment with HCl (Soil Survey Staff, 1990). The structure after acid treatment is fragile, however, and falls apart with handling. The laminar cap contains 742 g $\text{CaCO}_3 \text{ kg}^{-1}$, and the rest is mineral grains and opaline silica (Table 1). The highest $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio was obtained for this duripan in the Alko soil (Table 1). The SiO_2 content reported in Table 1 is based on 2.5-min boiling with 0.5 M NaOH and probably does not represent total noncrystalline and short-range-ordered SiO_2 in the samples. Observation of treated samples with the petrographic microscope showed that opaline silica was not removed completely with this brief 0.5 M NaOH treatment. The silica cement in the laminar cap is primarily opal-A, as indicated by the peak, centered near $4\text{\AA}$, in the XRD pattern (Fig. 4).

The thin, silica-rich layer from the lower part of the laminar cap does not contain nodules (bottom of Fig. 3B). The silica and CaCO_3 are closely associated and difficult to differentiate in the thin section. The BSE

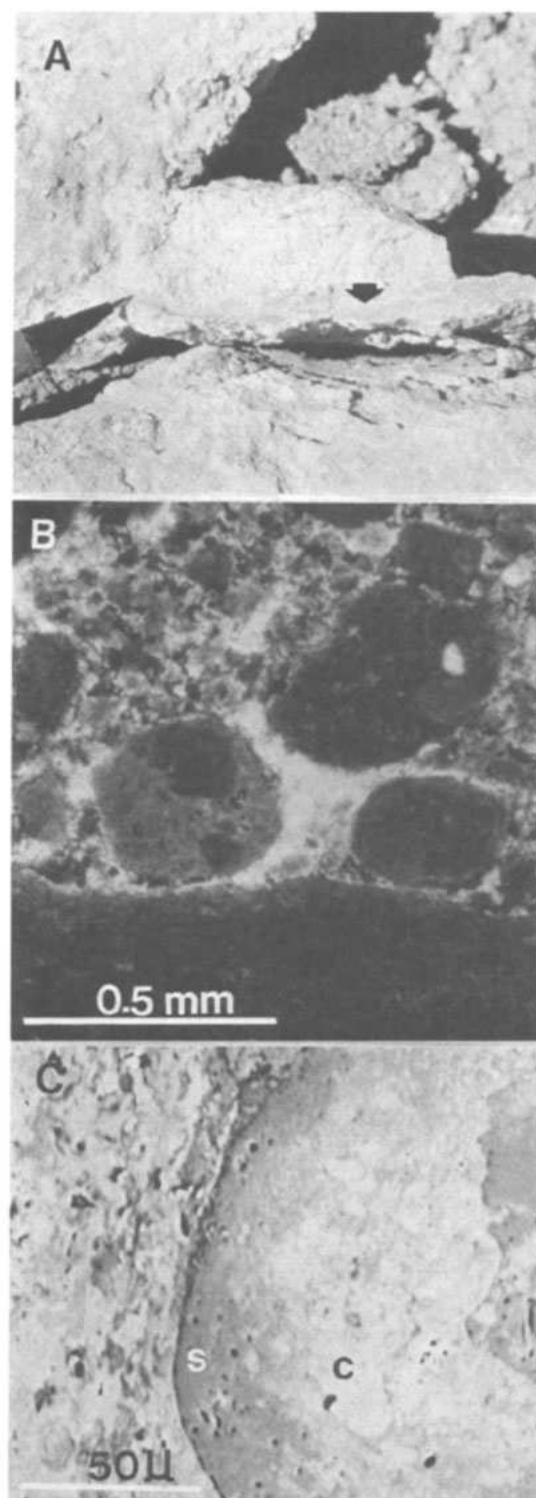


Fig. 3. (A) Laminar cap, 2 cm thick, with two distinct layers from the Alko soil. The arrow points to the silica-rich layer at the bottom; (B) photomicrograph of the laminar cap in a thin section (plane-polarized) showing the calcite-silica nodules on the top the silica-rich layer; (C) back-scattered electron photomicrograph of the lower layer of the laminar cap showing the areas occupied by CaCO_3 (c) and silica (s). The silica cement appears darker because of less electron scattering by the lighter Si.

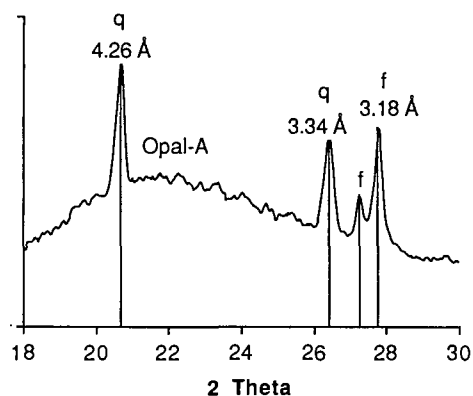


Fig. 4. The x-ray diffraction pattern of the opaline silica in the Alko laminar cap. Random powder mount in an aluminum holder. Cu K α radiation (q = quartz and f = feldspar).

micrograph, however, shows that CaCO₃ and opaline-silica do exist in separate areas (Fig. 3C). Discrete calcite crystallaria (Brewer, 1976), 3 to 5 μ m in diameter, occur in association with opaline silica. Boettinger and Southard (1990) also observed calcite as spherulitic crystallaria in the laminar caps of Durorthids on nearby granitic pediments of the western Mojave Desert.

The qualitative analysis by EDXRA of the nodules in the upper layer of the laminar cap shows that both CaCO₃ and silica are present (Table 2). These nodules are connected by silica bridges that are visible at their edges at high magnification (Fig. 5A). Grains of quartz and weathered feldspar are occasionally found in the center of the nodules (Fig. 5B). The feldspar grains are apparently weathering in situ and are one source for the silica cement (Boettinger and Southard, 1991; Southard et al., 1990).

Most of the morphological features of the laminar cap are preserved after removal of CaCO₃ (Fig. 6A), which shows that both calcite and silica are present in all of the structural features of the laminar cap. The nodules in the upper layer are not strongly bonded to the rest of the framework and some drop out during the treatment with HCl (Fig. 6B). Qualitative analysis of the nodules by EDXRA, after CaCO₃ removal, shows that silica is the major component of the nodule (Fig. 6B and Table 2). Chadwick et al. (1987) found a similar morphology in some sandy soils of Nevada, but the nodules in their

soils were only 10 to 150 μ m in diameter and consisted of nearly pure CaCO₃ imbedded in opaline silica void fillings.

Duripan Fragments on the Surface of the Alko Soil

Fragments of the laminar cap, 2 to 10 cm in diameter, cover 10 to 30% of the surface of the Alko soil. These fragments are similar to those found in the profile of the Alko, with a thick layer of spherical nodules and a thin silica-rich layer. Although the laminar duripan occurs at a depth of 10 cm, fragments are found at different depths in the upper 25 cm. The presence of krotovinas in the Alko soil suggests that duripan fragments are moved within the profile and to the surface. Some fragments have been on the surface for a long time, as is indicated by the lichen growth on their surfaces. They are normally thinner (0.5–1 cm) and less dense (1.70 Mg m⁻³) than the laminar caps in the profile (1–2 cm thick and 1.81 Mg m⁻³).

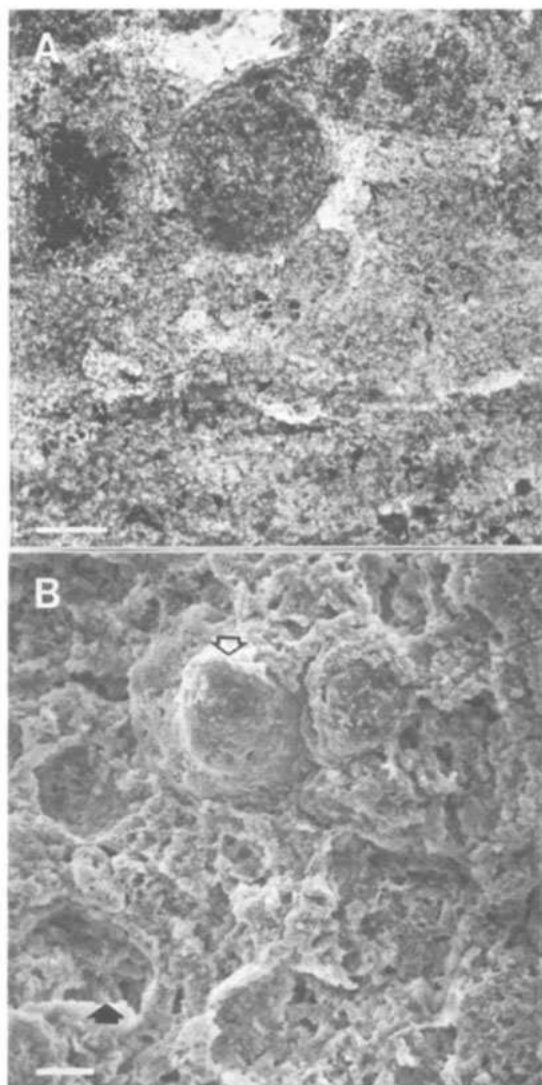


Fig. 5. (A) Thin-section photomicrograph (plane-polarized) of the nodules in the laminar cap of the Alko soil bridged by silica (arrows); (B) weathered, twinned feldspar grain in a spherical nodule (cross-polarized). The bars represent 100 μ m.

Table 2. Energy dispersive x-ray data for the micromorphological features of the cemented horizons of the Alko soil.

Figure	Micro-morphological features	Total counts in 20 s					
		Si	Ca	Al	Mg	K	Fe
		no.					
NS†	Nodules of the laminar cap (Bqkm)	2 842	16 500	—	—	—	—
6B	Nodules after carbonate removal	24 526	382	—	—	—	—
8C	Silica-clay bridging, Btqkm horizon	33 143	—	6319	337	1480	2248

† Micrograph not shown, but see Fig. 3B and 3C.

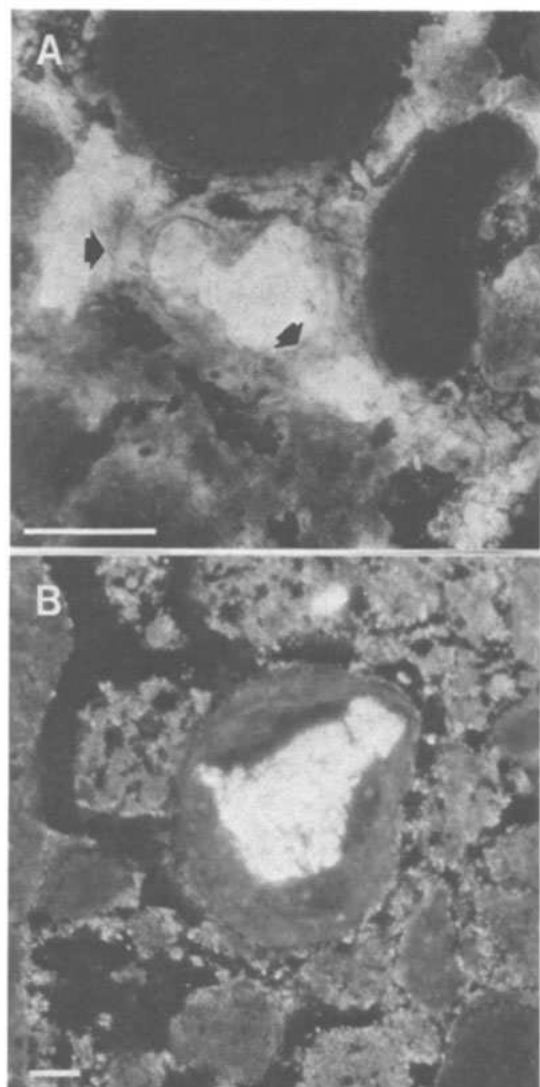


Fig. 6. (A) Photomicrograph of a thin section (plane-polarized) of the Alko laminar cap showing the silica framework after the removal of CaCO_3 cement. All features in the thin section become extinct when cross-polarized; (B) scanning electron micrograph of a nodule in the framework (open arrow) and outline (closed arrow) of nodules that have been plucked free during the HCl treatment. The bars represent $100\ \mu\text{m}$.

Surface duripan fragments become sources for enrichment of the profile by silica and carbonate through dissolution. The fragments are also susceptible to physical disintegration, producing small duripan fragments in the gravel fraction (Fig. 7A) and fragments of opaline silica in the coarse silt fraction of the surface (0–1 cm) horizon (Fig. 7B).

Lower Duripan in the Alko Soil

A second, morphologically different duripan (Btkm) is found at a depth of 40 to 150 cm in the Alko soil. This gravelly duripan has no laminar cap and is noncalcareous in the matrix. The cementation decreases with depth and laterally, downslope. Silica and clay occur as cutans (silans and argillans) and as grain bridges, 5 to $10\ \mu\text{m}$ thick (Fig. 8A, 8B, and 8C). The silica and clay occur side by side and overlap each other, indicating that

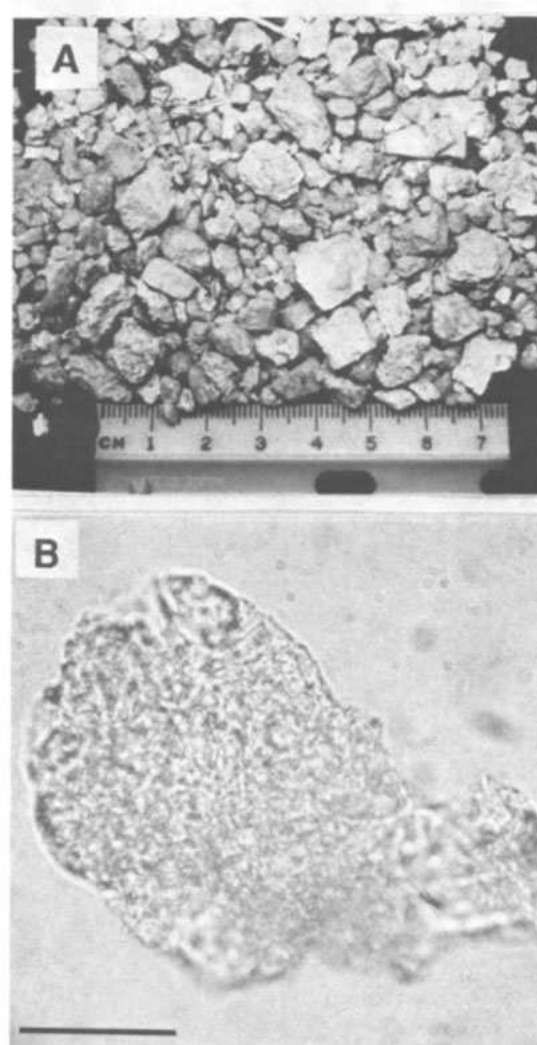


Fig. 7. (A) Small duripan fragments in the gravels from the surface of the Alko soil, products of physical disintegration of laminar duripans; (B) photomicrograph (plane-polarized) of an opaline-silica fragment found in the coarse silt fraction from the A1 horizon of the Alko soil. The bars represent $100\ \mu\text{m}$.

silica and clay illuviation probably occurred simultaneously. The clays are recognized by their reddish brown color and birefringence. Silica is isotropic and appears extinct in cross-polarized light. The clay layers are occasionally impregnated by silica and are difficult to differentiate from silica bridges and layers (bottom of Fig. 8A). The CaCO_3 is only present in seams, fractures, and old root channels; none was observed in the matrix. This suggests that the original duripan was carbonate-free and was subsequently calcified locally.

Sand and silt particles are cemented together, forming microagglomerates that are not dispersed in the fractionation process. Some of the microagglomerates in the very coarse sand fraction were observed with the scanning electron microscopy (Fig. 8C). The qualitative EDXRA analysis of the cement between the grains shows that, in addition to silica cement, clays are also present, as indicated by the presence of Al, K, Fe, and Mg (Table 2).

The weathering of the primary silicates could serve as

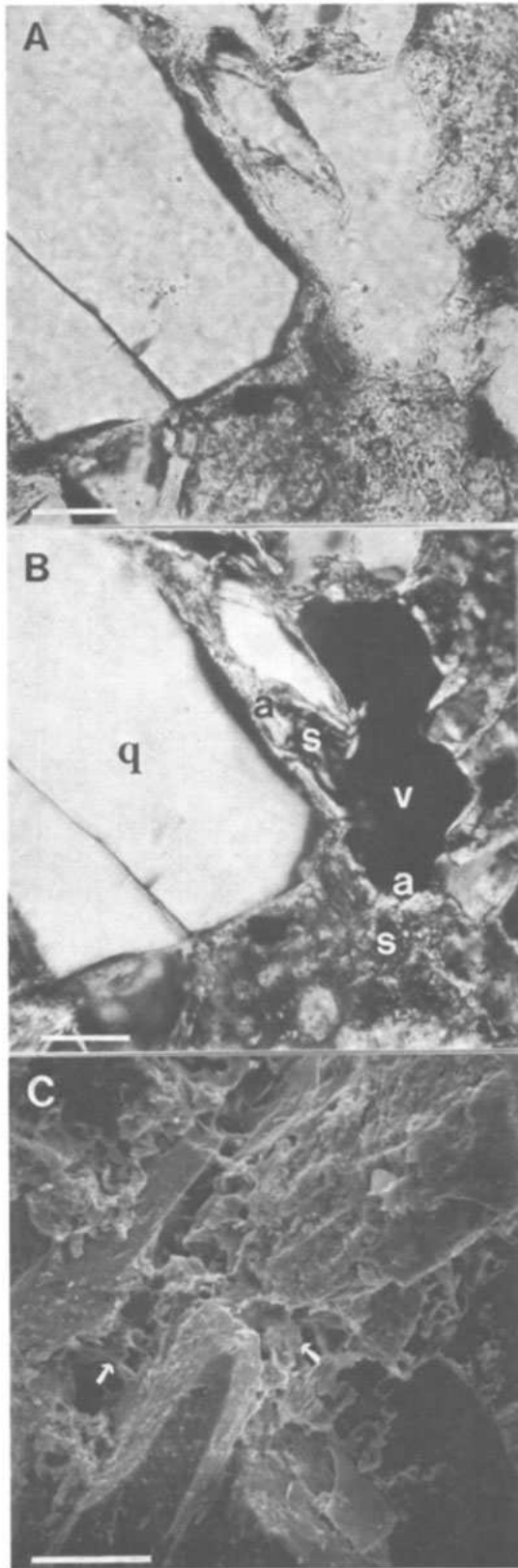


Fig. 8. Argillans, silans, and grain bridges in thin-section photomicrographs of the Btqkm horizon of the Alko soil in (A) plane-polarized and (B) cross-polarized light; (C) scanning electron photomicrograph of a microagglomerate from the very coarse sand fraction of the Btqkm horizon in the Alko soil. The grains are connected by bridges (arrows) of closely associated silica and clay (Table 2). The bars represent 100 μ m (a = argillan, s = silan, v = void, q = quartz).

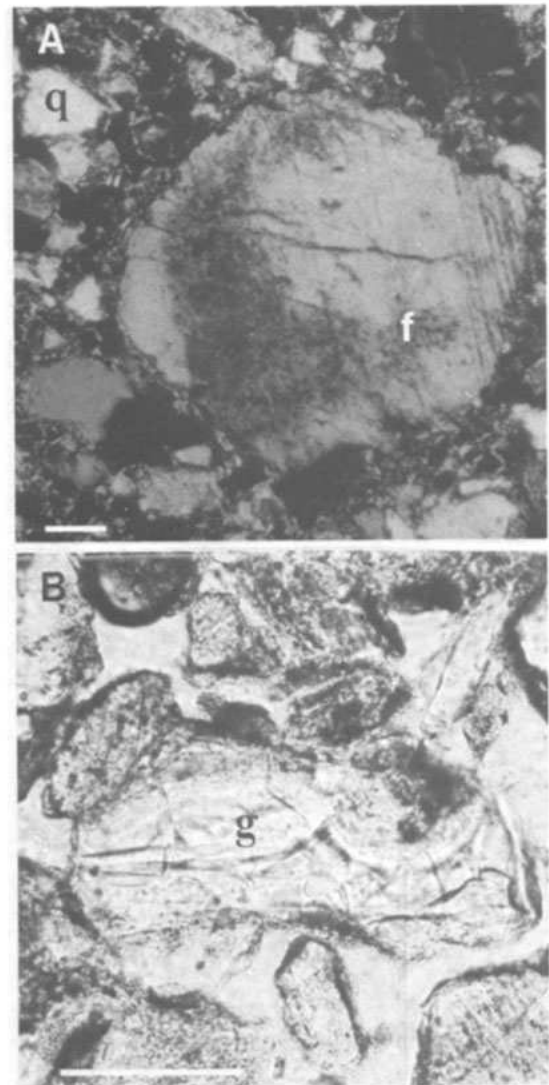


Fig. 9. (A) Thin-section photomicrograph of a feldspar grain (cross-polarized) that is relatively more weathered than quartz grains, and (B) volcanic glass (plane-polarized) in the Btqkm horizon of the Alko soil. The weathering of the feldspar is indicated by its turbidity, round shape, and etched edges. The bars represent 100 μ m (q = quartz, f = feldspar, g = glass).

the source for silica in this horizon. Boettinger and Southard (1991) have suggested that feldspars contribute silica to soils on the granitic pediments of the western Mojave Desert. Feldspars, especially plagioclase, are weathered relative to quartz in the lower duripan of Alko (Fig. 9A). The contribution of volcanic glass to the silica cement is probably less significant, since only a few grains of volcanic glass were found in the thin sections (Fig. 9B) and in the grain-count study of these soils (Eghbal and Southard, 1993b).

Based on paleomagnetic data, soil stratigraphy, and soil morphology, the lower duripan of the Alko soil probably formed 300 to 500 ka (Eghbal and Southard, 1993a). The dissolution of the glass shards should have occurred rapidly compared with crystalline silicates, especially during the moist periods of the Pleistocene. The presence and the preservation of the glass shards may

indicate that this horizon has been relatively isolated from soil-forming processes after its cementation by silica.

Duric Horizons in the Neuralia and Garlock Soils

Silica morphology and distribution in the Btqk horizon of the Neuralia soil differ from that in the Btqkm horizon of the Alko soil. In the Alko soil, skeletal grains are cemented by the silica (Fig. 8). In the Btqk horizon of the Neuralia, skeletal grains are not cemented; rather, much of the opaline silica occurs as isotropic flocs (Flach et al., 1974) in pores (Fig. 10A). Although opaline silica is only visible in pores, the Btqk horizons of the Neuralia soil exhibit partial cementation in the field, indicating silica also cements finer grains in the matrix. The clays occur as grain argillans and papules (Brewer, 1976). More clay is present in the matrix of the Btqk horizon of the Neuralia than in the Btqkm horizon of the Alko, as indicated by thicker argillans and the presence of papules. The presence of the silica in the pores suggests that much of the silica illuviation occurred subsequent to the clay illuviation in the gravelly horizons of the Neuralia.

Silica was observed in only a few pores in the Btqk horizon (135–150 cm) of the Garlock soil (Fig. 10B and 10C). This horizon, however, had one of the highest NaOH-extractable silica contents (41 g kg^{-1}) and a $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of 2.9 (Table 1). The clay content increased from 161 to 300 g kg^{-1} when the $<2\text{-mm}$ sample was treated with 0.5 M boiling NaOH to remove the non-crystalline and short-range-ordered aluminosilicates before the fractionation. The high NaOH extractable silica content and increase in clay content after the treatment indicate that silica is an important cementing agent in the Btqk horizon of the Garlock soil. The clay accumulations in this horizon show good orientation and are as much as $300 \text{ }\mu\text{m}$ thick (Fig. 10B and 10C).

As in the Alko soil, the contribution of volcanic glass to the silica cement in Btqk horizons of the Neuralia and Garlock soils is probably not significant. The influence of volcanic glass, however, cannot be completely discounted, since the absence of glass may mean it has been destroyed by weathering. The silicification of the Btqk horizons in the Neuralia soil apparently occurred after the formation of the argillic horizons, since silica occurs in pores. The silicification of the Btqk horizons in the Garlock soil may have occurred simultaneous with and after clay illuviation, as is indicated by the cementation of clay particles by silica and the presence of silica in pores. The weathering of silicate minerals such as feldspar (Fig. 10A) in these soils probably was the main source for the silica cement (Boettinger and Southard, 1991).

The distribution of the opaline silica in the Btqk horizons of the Neuralia and Garlock soils, and the lateral variation in the cementation of the lower duripan of the Alko, suggest that these soils have undergone different soil-forming processes. The lower duripan of the Alko soil and horizons below it have been practically isolated from downward percolating water and possibly root activity after the formation of the lower duripan, although localized weathering in the lower pan may occur via lateral water flow. The gravelly horizons of the Neuralia and Garlock are not completely cemented by silica, enabling CaCO_3 to move into and through the gravelly horizons to the underlying buried Bt horizon. The cal-

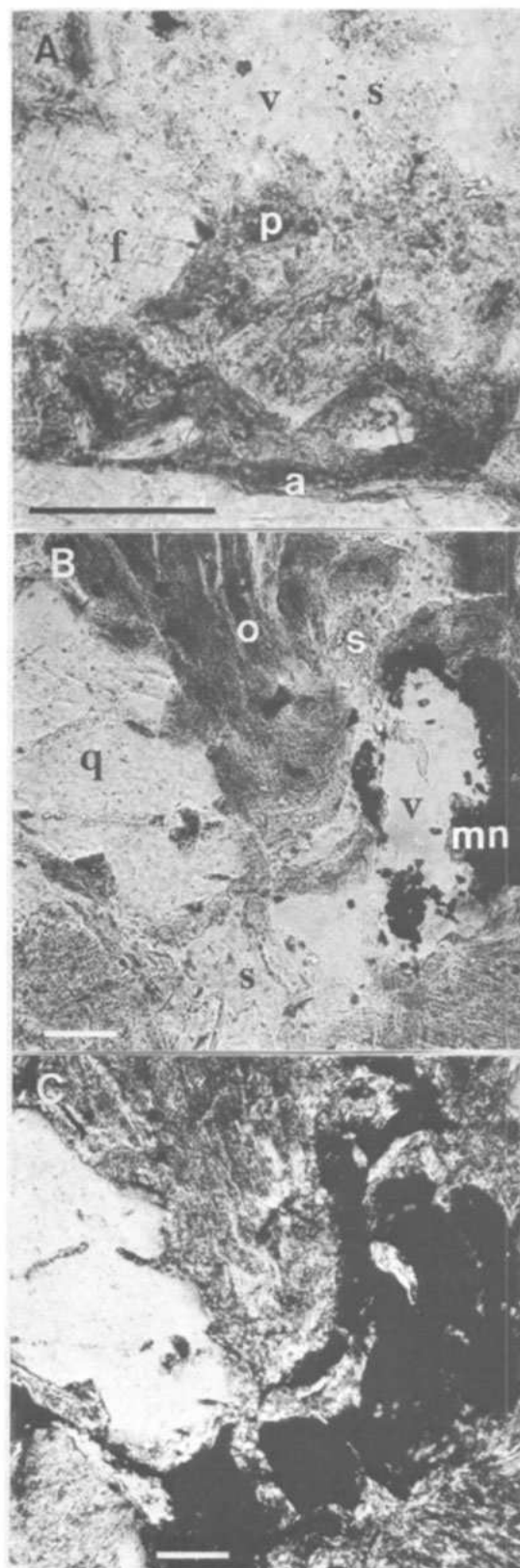


Fig. 10. Photomicrograph of thin sections from the Btqk horizons of the Neuralia (A, plane-polarized) and the Garlock soils (B, plane-polarized; C, cross-polarized). Silica flocs occur in the pores. The bars represent $100 \text{ }\mu\text{m}$ (a = argillan, p = papule, o = oriented clay in matrix, s = silica flocs, v = void, q = quartz, f = feldspar, mn = manganese concretions).

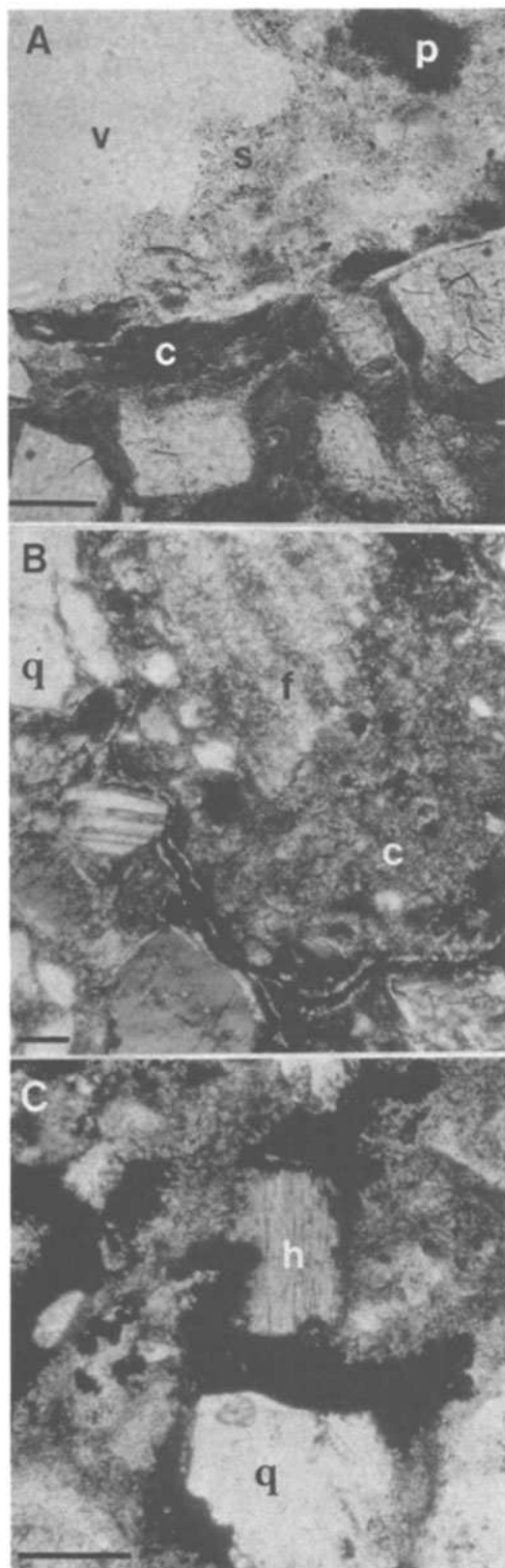


Fig. 11. Photomicrographs of a thin section of the buried Bt horizon in the Neuralia soil (A, plane-polarized). Silica flocs occur in the pores. Most of the feldspars (B, cross-polarized) and hornblende (C, plane-polarized) are weathered. The twinning of the weathered feldspar is barely visible and the bottom part is completely weathered to clay. The bars represent 100 μm (p = papule, c = clay, s = silica flocs, v = voids, q = quartz, f = feldspar, h = hornblende).

cification and much of the silicification of the Btqk horizons in these soils apparently occurred after the formation of the argillic horizons of the Neuralia and Garlock soils and the lower duripan in the Alko soil.

Buried Bt Horizons

The matrix of the buried Bt horizon is dominated by clays that seldom show orientation (Fig. 11A). The absence of oriented argillans throughout the Btb may indicate either that most clay deposits have formed in situ, or that orientation of argillans has been destroyed by calcification or other post-burial alteration. Silica occurs as flocs in the pores (Fig. 11A).

Many of the feldspars and hornblende grains are highly weathered, and some of them are recognizable only by their shape and weathering products (Fig. 11B and 11C). The weathered hornblendes usually have an opaque alteration rim; feldspars apparently weather to clays. The high birefringence in some weathered feldspars may indicate that calcite is replacing the feldspar (not shown in Fig. 11).

Although the total carbonate content of the buried Bt horizon increases from the Alko to the Garlock, the morphology of carbonate distribution remains similar. The ped faces are coated with CaCO_3 , but none occurs inside the peds, suggesting that the calcification occurred after the Bt horizon formed. The increase in the CaCO_3 content of the buried soil, from 2 g kg^{-1} under the Alko to 11 and 66 g kg^{-1} under the Neuralia and Garlock, respectively (Table 1), indicates that calcification of the buried Bt started before the cementation of the lower duripan of the Alko. Calcification below the Alko subsequently ceased when its duripan became cemented, but has advanced further in the Neuralia and Garlock soils. Similarly, the cementation of the buried Bt horizon by silica increases from the Alko to the Neuralia and Garlock, as indicated by the increase in the NaOH-extractable SiO_2 and the $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio (Table 1). Higher CaCO_3 and silica contents in the buried Bt horizons of the Neuralia and Garlock soils indicate that water could move through these soils without duripans and reach the buried Bt horizons during wetter years, possibly during Pleistocene pluvials. Due to complete cementation, carbonates and silica are unlikely to leach through the lower duripan of the Alko, no matter how wet the season is. The increase in the CaCO_3 and silica contents in the buried Bt from the Alko to Garlock soil could also be due to lateral flow during wet periods.

Sequence of Pedogenic Events

It is possible to propose the following sequence of pedogenic events based on soil stratigraphy and clay, silica, and CaCO_3 distribution in these soils. (i) Magnetic polarity of the buried Bt horizon, underlying all three soils, indicates formation prior to 783 ka. (ii) The calcification of the buried Bt horizon probably started before silica cementation of the lower duripan in the Alko soil reduced downward percolation of water, and may have occurred before deposition of Unit II (Fig. 2). The lower duripan formed within the past 783 ka, with simultaneous silicification and clay illuviation, and was subsequently calcified, as is indicated by the occurrence of CaCO_3 only in seams, fractures, and root channels. (iii)

The argillic horizon in the Neuralia soil, Unit III, probably formed ≈ 200 to 300 ka (Eghbal and Southard, 1993a). Silica and, subsequently, CaCO_3 were deposited in the Btqk horizon of the Neuralia. Silicification and calcification of the buried Bt horizon probably started prior to deposition of Unit III, but advanced further during the formation of the Neuralia soil. At the same time, silica and CaCO_3 started to accumulate above the lower duripan in the Alko soil. (iv) Carbonates were leached and the argillic horizon formed in the Garlock soil of Unit IV, probably in the past 100 ka (Eghbal and Southard, 1993a). Silica and CaCO_3 further accumulated in the Btqk and the buried Bt horizon in the Garlock soil. (v) Duripan fragments and calcareous soil material were brought to the surface of the Alko and Neuralia soils by rodents, providing a local source for calcification of these two soils. The upper 65 cm of the Garlock soil has not been significantly recalcified.

Rodents apparently played an important role in the evolution of the Alko and Neuralia soils by creating a younger, pedogenically rejuvenated, geomorphic surface through mixing of the upper soil mantle and addition of calcareous material to the surface. The upper 40 to 65 cm of these two soils can be viewed as forming from calcareous parent material with constant surficial addition of silica and CaCO_3 by the bioturbation process. This rejuvenation is restricted to the upper soil mantle of the Alko and Neuralia soils, above the Btqk(m) horizons, where rodents apparently have an easier time digging through the softer calcareous material. Such bioturbation may be limited in the Garlock soil by the inability of rodents to dig in the hard, sandy clay loam of the argillic horizon.

It is tempting to attribute the complicated polygenesis of soils forming in the study area to climatic fluctuations. Reheis (1897), for example, speculated that laminated argillans and calcans in semiarid soils of Montana (Argiborolls?) were related to local glacial-interglacial episodes throughout the Quaternary. From packrat midden contents and climatic modeling, Spaulding and Graumlich (1986) concluded that, during the last full glacial (18 ka), the Mojave Desert south of 36°N had a climate with wetter winters and cooler temperatures than the modern climate. They estimated the temperature to have been at least 6°C cooler, and precipitation to have been no more than 40% greater, than present. Even with these conditions, desert shrub communities apparently occurred at elevations below 1000 m (Spaulding and Graumlich, 1986), suggesting that pluvial effects may have been minimal at low elevations. McFadden et al. (1986) attributed late-Pleistocene episodic loess deposition and soil formation in the Cima volcanic field of the eastern Mojave Desert to increasing aridity during interpluvial times and promotion of soil development during the more humid pluvial periods.

Details of climatic conditions in the Mojave Desert earlier in the Pleistocene are not well known, but it is not unreasonable to assume that cooler temperatures during glacial episodes gave rise to reduced evapotranspiration rates. Southward displacement of the winter storm track (Spaulding and Graumlich, 1986) during glacial periods may have increased winter precipitation as well. We speculate that the noncalcareous soils, with well-developed argillic horizons, developed during pluvial pe-

riods related to Sierran glaciations, due to greater effective moisture. The buried soil, before truncation and burial, the lower duripan in the Alko soil, and the argillic horizons of the Garlock and Neuralia soils probably formed, for the most part, under such climates. On the other hand, calcification of the buried soil, the duric horizons in the Neuralia and Garlock, and the upper portion of the Alko soil probably occurred during drier interpluvial times. Silicification appears to have been contemporaneous with both clay illuviation (lower Alko duripan and Garlock Btqk) and calcification (Neuralia Btqk and upper Alko duripan), suggesting that silicification may be related more to pedochemical conditions (Munk and Southard, 1993) than to climatic conditions.

Some pedogenic processes such as bioturbation and the addition of calcareous materials to the surfaces of the Alko and Neuralia soils may not be directly related to climatic events. The impact of bioturbation during the present-day dry climate is that silica and carbonates are not completely leached from the surface horizons of these two soils.

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